

By starting at the retirement problem and working backward, life-cycle models make clear that the true risk-free asset for retirees is not T-bills. Risk-free assets are securities that can lock-in increasing retiree spending in real terms.⁴² The only way to do this is to hold a portfolio of real bonds (TIPS, see chapter 11) that completely *immunizes* a retiree's cash flow needs at retirement. Real bonds, however, have prices that are quite volatile and funding these retirement needs with real bonds can open up the investor to substantial investment risk as real rates change over time. Today's negative real yields mean that an individual, instead of counting on bond earnings to help achieve the savings goal, has to set aside even more money than will be needed later in retirement.

4.2. ARE WE SAVING ENOUGH?

Using life-cycle models, Skinner (2007) estimates how much we should save for a comfortable retirement. The numbers are startlingly high. Consider a forty-year old planning to retire at sixty-five, living to ninety-five, saving 7.5% of pre-tax earnings, and hoping income in retirement is just 30% of pre-retirement earnings. At age forty, this individual needs to have a wealth-to-income ratio of approximately 2, rising to 3 at age fifty and peaking at 5 when retirement occurs. This wealth does not include housing, and for non-homeowners the numbers are even larger at 4, 5, and 8 for ages forty, fifty, and sixty-five, respectively.

Most people are nowhere near these levels of savings.

Most households fail to save the amounts required by life-cycle models to ensure a comfortable retirement. Lusardi and Mitchell (2006) report that the early baby boomers had a median wealth-to-income ratio of just 2, far less than the optimal savings numbers calculated by Skinner (2007). Poterba, Venti, and Wise (2011a) report that half of households headed by someone sixty-five to sixty-nine had less than \$52,000 in total net wealth in 2008, and 43% of such households could not scrounge up \$25,000—which is often the minimum investment in annuities (see below). According to the Employee Benefit Research Institute in 2013, 57% of U.S. workers reported less than \$25,000 in total savings excluding their homes, and 28%—a clear-eyed minority—were not confident they had enough money to retire comfortably.⁴³ An amazing 34% report that they had not saved anything for retirement!

The first-order problem is getting people to save. Consumption today is always more appealing than saving for tomorrow. Economists have modeled this type of behavior using a *hyperbolic discounting* function, which is the St. Augustine version

⁴² See Bodie (1990b) and Campbell and Viceira (2001).

⁴³ The 2013 Retirement Confidence Survey: Perceived Saving Needs Outpace Reality for Many, Employee Benefit Research Institute, Issue Brief, March 2013, No. 384.